

Examples of ionising radiation include alpha particle radiation, beta particle radiation, neutrons, gamma rays and X-rays.

Examples of non-ionising radiation are sound waves, visible light, RF and microwaves. The RF spectrum covers the range of about 3 kHz to 300 GHz. The cellphone frequencies are the sets of frequency ranges within the UHF band (300 MHz and 3 GHz or 3000 MHz) of RF spectrum. Non-ionising radiation is considered to be harmless below the levels that cause heating. Higher levels of non-ionising radiation are dangerous in direct exposure, although the degree of danger is a subject of debate.

Concept of EM radiation from an antenna. Most of us are aware that a current-carrying conductor is associated with a magnetic field. As current in the conductor changes, the magnetic field also changes, and a changing magnetic field produces a changing electric field around itself. The changing electric field is again associated with a changing magnetic field. Thus the two fields are self-sustaining. The phenomenon is shown in Fig. 2.

To understand how electromagnetic radiation takes place, consider an antenna (say, a dipole) to be a capacitor whose plates have been taken apart. The associated EM fields are called induction fields, which need a conductor (here, it is antenna) to support them. At low frequencies, the collapsing E and H fields return all the energy supplied to the induction field to the antenna. No energy from the induction field is radiated from the antenna. The EM fields associated with the antenna vary when the generator voltage goes through one complete cycle (360°). Corresponding H- and E-field amplitude variations on time scale are shown in Fig. 3.

When the frequency of AC voltage source is made sufficiently high, the induction field is produced as described above. However, the generator voltage reverses direction before the electric field has had time to collapse

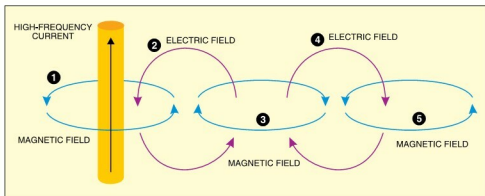


Fig. 2: EM field surrounding a conductor carrying varying current

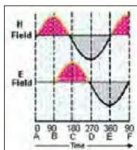


Fig. 3: Time-varying H and E fields around antenna

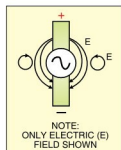


Fig. 4: Repulsion of collapsing E field

with its H-field counterpart) into space as depicted in Fig. 4. (Note that for brevity, only E field is shown; the accompanying H field is not shown.)

Phase relationships between H- and E-field components of induction and radiation fields along with direction of their propagation are shown in Figs 5 and 6, respectively.

Polarisation. The orientation of an electric field vector in the plane orthogonal to the direction of propagation is called polarisation. If the electric field vector is always oriented in a given direction, the wave is linearly (either horizontally or vertically) polarised. If the electric field vector rotates around the direction of propagation, maintaining a constant magnitude, the wave is circularly polarised. If the extremity of the electric field vector traces an ellipse, the wave is elliptically polarised. The electric field vector rotates either clockwise or counter-clockwise.

It is difficult to predict the orientation of the electric field in the near-field region as the transmitting antenna cannot be considered as a point source in this region. However, in the far-field region, the antenna behaves like a point source, and the electric and magnetic components of the field become orthogonal to the direction of propagation and their polarisation characteristics do not vary with distance.

EM field regions surrounding an antenna. The space surrounding an antenna is generally subdivided into three regions—reactive near-field,

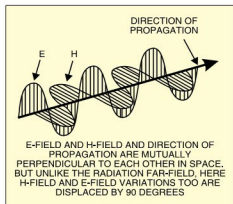


Fig. 5: Induction field components

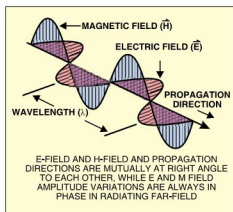


Fig. 6: Radiation field components

completely. The direction of new electric lines of force is such as to repel the previous collapsing electric field (along